**PROBLEM SET 2, PART 2: REGULAR VALUES**
**DUE: OCT. 17**

(1) [**Measure zero set in smooth manifolds**]
   (a) Prove: the phrase "measure zero" is well-defined on smooth manifolds.
   (b) Deduce Sard's theorem from the Euclidean case.
   (c) Show that if $f : M \to N$ is a smooth map of constant rank $r < \dim N$, then $f(M)$ has measure zero.

(2) [**An counterexample to Sard's Theorem**]
Here is a counterexample to Sard's theorem if $f$ is not smooth enough (constructed by E. Grinberg). Let $C \subset [0,1]$ be the Cantor set.
   (a) Construct a $C^1$ function $f : \mathbb{R} \to \mathbb{R}$ such that the critical set of $f$ contains $C$. (Hint: Denote $[0,1] \setminus C = \cup(a_k, b_k)$. Start with a "continuous bump function" $f_k$ on $(a_k, b_k)$ with $\int f_k(t)dt = b_k - a_k$.)
   (b) Show that the function $g : \mathbb{R}^2 \to \mathbb{R}$ defined by $g(x, y) = f(x) + f(y)$ is $C^1$, and the set of critical values contains an interval. (Hint: Show that $C + C = [0, 2]$.)

(3) [**Morse functions**]
Let $U \subset \mathbb{R}^n$ be an open set, and $f \in C^\infty(U)$.
- We say a critical point $p \in U$ of $f$ is *non-degenerate* if the Hessian matrix

$$\mathrm{Hess} f(p) = \left( \frac{\partial^2 f}{\partial x^i \partial x^j} \right)(p)$$

  is non-degenerate.
- A function is called a *Morse function* if every critical point is non-degenerate.

Prove:
   (a) Use inverse function theorem to prove that non-degenerate critical point must be isolated.
   (b) Given any $f \in C^\infty(U)$, for almost every $a \in \mathbb{R}^n$, the "linear perturbation"

$$f_a : U \to \mathbb{R}, \quad x \mapsto f(x) + a_1 x^1 + \cdots + a_n x^n$$

   of $f$ is a Morse function on $U$.
   [Hint: Consider regular values of the map $g = df = (\frac{\partial f}{\partial x^1}, \cdots, \frac{\partial f}{\partial x^n}) : U \to \mathbb{R}^n$.]
   (c) (Not required) Suppose $U$ is bounded. Prove: for any $f \in C^\infty(U)$ and any $\varepsilon > 0$, there is a Morse function $g \in C^\infty(U)$ so that $|g - f| < \varepsilon$ and all critical values of $g$ are distinct.
   (d) (Not required) Extend the result in (c) to smooth functions defined on a compact manifold.

(4) [**The Lagrange multiplier**]
Let $M$ be a smooth manifold, and $f \in C^\infty(M)$ a smooth function. We would like to study the critical points of the function $\tilde{f} := f|_S \in C^\infty(S)$ for a smooth submanifold $S \subset M$. For simplicity, we suppose there is a smooth map $g : M \to N$ and a regular value $p \in N$ of $g$ so that $S = g^{-1}(q)$. Prove: a point $p \in S$ is a critical point of $\tilde{f}$ if there exists a linear function $L : T_qN \to \mathbb{R}$, (called a *Lagrange multiplier*), so that $df_p = L \circ dg_p$.

(5) [**Proper maps**]
Recall that a map is called *proper* if the pre-image of any compact set is compact. Let $f : M \to N$ be a smooth and proper map.
(a) Prove: If an injective immersion $f : M \to N$ is proper, then it is an embedding.
(b) Now suppose $\dim M = \dim N$, and suppose $q \in f(M)$ be a regular value of $f$. Prove: $f^{-1}(q)$ is a finite set $\{p_1, \cdots, p_k\}$, and there exist a neighborhood $V$ of $q$ in $N$ and neighborhoods $U_i$ of $p_i$ in $M$ such that
- $U_1, \cdots, U_k$ are disjoint coordinate charts in $M$,
- $f^{-1}(V) = U_1 \cup \cdots \cup U_k$,
- For each $1 \leq i \leq k$, $f$ is a diffeomorphism from $U_i$ onto $V$.

(6) [**The cotangent bundle**]
Let $M$ be a smooth manifold of dimension $n$. Let $T_p^*M$ be the dual vector space of $T_pM$, with a dual basis $\{dx^1, \cdots, dx^n\}$ (which is defined locally for a coordinate chart of $M$) which is defined to be the dual of $\{\partial_1, \cdots, \partial_n\}$. Let $T^*M = \bigcup_p T_p^*M$ be the disjoint union of all $T_p^*M$. We will call $T^*M$ the *cotangent bundle* of $M$.
(a) Modify PSet2-1-3 to endow with $T^*M$ a topology so that it is a smooth manifold of dimension $2n$.
(b) Prove: $T^*M$ is orientable.
(c) (Not required) Prove: If $f$ is a smooth function on $M$, then the map
$$s_f : M \to T^*M, \quad p \mapsto (p, df_p)$$
is an injective immersion and is proper. [In particular, its image is a smooth submanifold of $T^*M$.]
(d) (Not required) For any $(p, \xi_p) \in T^*M$, the tangent space $T_{(p,\xi_p)}T^*M \simeq T_pM \oplus T_p^*M$.